浙江大学2015-16秋冬《数据结构基础》期中模拟练习

开始时间	1/1/2016, 12:00:00 AM	**结束时间**	1/18/2038, 12:00:00 AM	**答题时长**	90分钟
考生	–	**得分**	95	**总分**	100

判断题

总分：10　得分：8

1-1 If the postorder and inorder traversal sequences of a binary tree are the same, then none of the nodes in the tree has a right child. (2分)

◉ T　○ F

评测结果：答案正确（2 分）

1-2 $N(logN)^2$ is $O(N^2)$. (2分)

○ T　◉ F

评测结果：答案错误（0 分）

1-3 If the most commonly used operations are to visit a random position and to insert and delete the last element in a linear list, then sequential storage works the fastest. (2分)

◉ T　○ F

评测结果：答案正确（2 分）

1-4 Given the input sequence onto a stack as {1, 2, 3, ..., N}. If the first output is i, then the j-th output must be $j-i-1$. (2分)

○ T　◉ F

评测结果：答案正确（2 分）

1-5 The best "worst-case time complexity" for any algorithm that sorts by comparisons only must be $O(NlogN)$. (2分)

◉ T　○ F

评测结果：答案正确（2 分）

单选题

总分：64　得分：61

2-1 For a binary search tree, in which order of traversal that we can obtain a non-decreasing sequence? (3分)

- A. preorder traversal
- B. postorder traversal
- C. inorder traversal
- D. level-order traversal

评测结果：答案正确（3 分）

2-2 Which one of the following relations is correct about the extra space taken by heap sort, quick sort and merge sort? (3分)

- A. heap sort ＜ merge sort ＜ quick sort
- B. heap sort ＞ merge sort ＞ quick sort
- C. heap sort ＜ quick sort ＜ merge sort
- D. heap sort ＞ quick sort ＞ merge sort

评测结果：答案正确（3 分）

2-3 Given the pushing sequence of a stack as {1, 2, 3, 4, 5}. If the first number being popped out is 4, then the last one out must be: (3分)

- A. 1
- B. 3
- C. 5
- D. 1 or 5

评测结果：答案正确（3 分）

2-4 Insert {5, 2, 7, 3, 4, 1, 6} one by one into an initially empty min-heap. The preorder traversal sequence of the resulting tree is: (6分)

- A. 1, 3, 2, 5, 4, 7, 6
- B. 1, 2, 3, 4, 5, 7, 6
- C. 1, 2, 5, 3, 4, 7, 6
- D. 1, 3, 5, 4, 2, 7, 6

评测结果：答案正确（6 分）

2-5 Given a tree of degree 4. Suppose that the numbers of nodes of degrees 2, 3 and 4 are 4, 2 and 1, respectively. Then the number of leaf nodes must be: (6分)

- A. 8
- B. 12
- C. 18
- D. 20

评测结果：答案正确（6 分）

2-6 Suppose that the height of a binary tree is h (the height of a leaf node is defined to be 1), and it has only the nodes of degrees 0 and 2. Then the minimum and maximum possible total numbers of nodes are: (6分)

- A. $2h, 2^h - 1$
- B. $2h - 1, 2^h - 1$
- C. $2h - 1, 2^{h-1} - 1$
- D. $2^{h-1} + 1, 2^h - 1$

评测结果：答案正确（6 分）

2-7 Suppose that an array of size `m` is used to store a circular queue. If the front position is `front` and the current size is `size`, then the rear element must be at: (6分)

- A. `front+size`
- B. `front+size-1`
- C. `(front+size)%m`
- D. `(front+size-1)%m`

评测结果：答案正确（6 分）

2-8 Suppose that the level-order traversal sequence of a min-heap is {1, 3, 2, 5, 4, 7, 6}. Use the linear algorithm to adjust this min-heap into a max-heap. The inorder traversal sequence of the resulting tree is: (6分)

- A. 3, 5, 4, 2, 6, 1, 7
- B. 1, 4, 3, 7, 2, 6, 5
- C. 3, 5, 4, 7, 2, 6, 1
- D. 4, 1, 3, 7, 6, 2, 5

评测结果：答案正确（6 分）

2-9 Among the following sorting methods, which one may cause that none of the elements are at their final positions before the last run begins? Assume that there are more than 2 elements to be sorted. (3分)

- A. bubble sort
- B. insertion sort
- C. heap sort
- D. quick sort

评测结果：答案错误（0 分）

2-10 To delete `p` from a doubly linked list, we must do: (6分)

- A. `p->prior=p->prior->prior; p->prior->next=p;`
- B. `p->next->prior=p; p->next=p->next->next;`
- C. `p->prior->next=p->next; p->next->prior=p->prior;`
- D. `p->next=p->prior->prior; p->prior=p->next->next;`

评测结果：答案正确（6 分）

2-11 Given input { 321, 156, 57, 46, 28, 7, 331, 33, 34, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort? (4分)

- A. →331→321→33→63→34→156→46→57→7→28
- B. →321→331→33→63→34→156→46→57→7→28
- C. →156→28→321→331→33→34→46→57→63→7
- D. →57→46→28→7→33→34→63→156→321→331

评测结果：答案正确（4 分）

2-12 The array representation of a disjoint set containing numbers 0 to 8 is given by { 1, -4, 1, 1, -3, 4, 4, 8, -2 }. Then to union the two sets which contain 6 and 8 (with union-by-size), the index of the resulting root and the value stored at the root are: (6分)

- A. 1 and -6
- B. 4 and -5
- C. 8 and -5
- D. 8 and -6

评测结果：答案正确（6 分）

2-13 If on the 9th level of a complete binary tree (assume that the root is on the 1st level) there are 100 leaf nodes, then the maximum number of nodes of this tree must be: (6分)

- A. 311
- B. 823
- C. 1847
- D. cannot be determined

评测结果：答案正确（6 分）

程序填空题

总分：6　得分：6

5-1 The function is to lower the value of the integer key at position `P` by a positive amount `D` in a min-heap `H`.

```
void DecreaseKey( int P, int D, PriorityQueue H )
{
   int i, key;
   key = H->Elements[P] - D;
   for ( i = P                                (3分); H->Elements[i/2] > key; i/=2
       H->Elements[i]=H->Elements[i/2]        (3分);
   H->Elements[i] = key;
}
```

评测结果：答案正确（6 分）

序号	结果	得分
0	答案正确	3
1	答案正确	3

函数题

总分：20　得分：20

6-1 No Less Than X in BST

You are supposed to output, in decreasing order, all the elements no less than `X` in a binary search tree `T`.

Format of function:

```
void Print_NLT( Tree T,  int X );
```

where `Tree` is defined as the following:

```
typedef struct TreeNode *Tree;
struct TreeNode {
```

```
    int Element;
    Tree  Left;
    Tree  Right;
};
```

The function is supposed to use `Output(X)` to print `X`.

Sample program of judge:

```
#include <stdio.h>
#include <stdlib.h>

typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree  Left;
    Tree  Right;
};

Tree BuildTree(); /* details omitted */
void Output( int X ); /* details omitted */

void Print_NLT( Tree T,  int X );

int main()
{
    Tree T;
    int X;

    T = BuildTree();
    scanf("%d", &X);
    Print_NLT( T, X );
    printf("End\n");

    return 0;
}

/* Your function will be put here */
```

Sample Output 1 (for the tree shown in Figure 1):

```
92 91 90 85 81 80 End
```

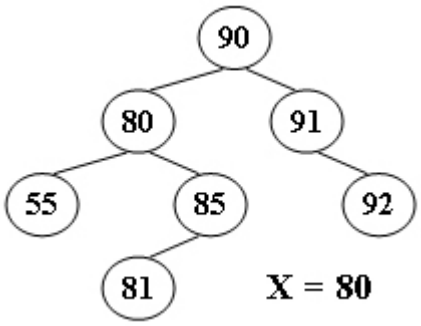


Figure 1

Sample Output 2 (for the tree shown in Figure 2):

```
End
```

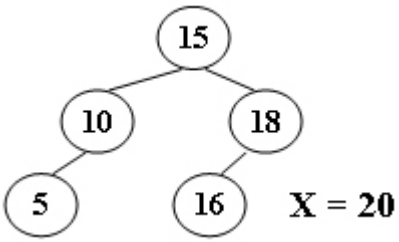


Figure 2

代码

```
int a[1000] = {0}, st = -1;

int cmp(const void *l, const void *r) {
    return *(int*)r-*(int*)l;
}

void Tr(Tree T, int X) {
    if (T) {
        if (T->Element >= X) {
            a[++st] = T->Element;
        }
        Tr(T->Left, X);
        Tr(T->Right, X);
    }
}

void Print_NLT( Tree T,  int X ) {
    Tr(T, X);

    qsort(a, st+1, sizeof(int), cmp);
    for (int i = 0; i<=st; i++) {
        Output(a[i]);
    }
}
```

评测结果：答案正确（20 分）

测试点	结果	得分	耗时	内存
0	答案正确	7	2 ms	256 KB
1	答案正确	3	2 ms	256 KB
2	答案正确	6	2 ms	296 KB
3	答案正确	2	2 ms	256 KB
4	答案正确	2	3 ms	256 KB

```
a.c: In function ‘BuildTree’:
a.c:33:6: warning: ignoring return value of ‘scanf’, declared with attribute warn_u
      scanf("%d", &n);
      ^~~~~~~~~~~~~~~
a.c:35:10: warning: ignoring return value of ‘scanf’, declared with attribute warn_
          scanf("%d", &x);
          ^~~~~~~~~~~~~~~
a.c: In function ‘main’:
a.c:52:5: warning: ignoring return value of ‘scanf’, declared with attribute warn_u
     scanf("%d", &X);
     ^~~~~~~~~~~~~~~
```